

Unpack a Suitcase SOP

One episode empties one suitcase into its zones. The zip is drawn open, the lid falls back onto its rest, three folded garments go onto three spots sorted by type, the toiletries go into the toiletry caddy, the pouch comes out onto the pouch spot, both shoes come out of the tote and onto the mat, and the pouch is nested inside the tote. The tote never leaves the case: it is worked where it stands, and the episode ends with the case holding nothing but the tote with the pouch inside it and the lid still open.

The order never changes: zip, lid, garments, toiletries, pouch, shoes, nest. A zone is finished before the next one starts, and once a step is over nothing it placed is touched again.

The zip has two sliders parked together in the middle of the track, and each gripper takes the one on its own side. The right slider is drawn out to the right end stop first, with the left gripper holding the case down, and then the left slider is drawn out to the left end stop with the right gripper holding it down. A draw takes as many pulls as the arm's travel needs: the tab is taken again and pulled again, always onward toward the end stop, until the slider will not travel further.

Everything else is split. The right gripper opens the lid on its own, moves every garment, and takes the right shoe. The left gripper stands off and does nothing while the lid is opened, then moves the toiletries, the pouch, and the left shoe. The tote's mouth is the one thing both arms take in turn: whichever gripper is not lifting a shoe or the pouch holds the mouth steady for the one that is. No garment ever touches the left gripper and the right gripper never reaches the toiletry caddy or the pouch spot.

The case is never lifted or moved, the tote is never lifted out of it, nothing is turned in the air, only the top item of a stack is ever taken, and no gripper opens anywhere but over the place the item is going.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual
- **parked_side:** none

Hardware checklist

1. Cameras are on and recording.

2. The environment camera shows the whole tabletop: the case lying flat at the back, the three garment spots down the right, the toiletry caddy and the pouch spot down the left, and the shoe mat at the front center.
3. The whole zip track is in frame end to end, the case interior is visible from above so every lift can be seen leaving it, the mouth of the tote is visible from above so the shoes can be seen coming out of it, and the packed contents are visible from the side so which item is on top of a stack can be seen.
4. Both arms are at home with grippers open.
5. The tabletop is clear of anything but the zones listed below.
6. Nothing obstructs the paths between the case and the four destination zones. Each arm reaches the side panel on its own side of the case, the top face of the closed lid on its own side, the middle of the zip track where the sliders park, and the end stop on its own side. Each arm reaches the middle of the tote's mouth rim and its own half of the shoe mat.
7. The right arm reaches the lid's front right corner and can carry it back over the hinge past upright. It does not have to reach the lid rest: the lid is released once it is past upright and falls the rest of the way onto the rest on its own.

Materials checklist

1. One suitcase lying flat at the back of the tabletop, base down on the tabletop and squared to the front edge, lid closed and facing up. Nothing holds the case but the tabletop it lies on.
2. The zip is a single straight track running left to right across the front edge of the case, with **two sliders parked together at the middle of the track**, one facing each end stop. Each slider carries a fabric loop tab big enough for a gripper to pinch, and the two tabs lie clear of each other so a gripper can close on one without touching the other.
3. The lid is hinged along the back edge and falls back onto the lid rest once it is carried past upright. Its **front right corner** is a firm edge a gripper can close across from above, with nothing packed against it and no trim or handle in the way. Its **top face is flat and firm**, clear of the zip track, so a gripper can press down on it while the case is still closed.
4. The case is small, so it is not packed in one layer: some items are **stacked** on top of others. Every stack is packed in the order it is worked, so the item that comes out next is always the one lying on top and nothing ever has to be dug out from under anything. Stacks stand clear of one another so a gripper can close on the top item of one without touching the next.
5. The right half of the case holds one **garment stack**: three folded garments, one of each type, each folded face up into a flat rectangle whose **left side** is a clean folded edge a gripper can close across. They are stacked in working order, the **t-shirt on top**, the **pants** under it, and the **shorts** on the case floor, squared up so no garment overhangs the one below it.
6. The left half of the case holds the **pouch lying flat on the case floor**, empty and square to the case's left wall, with the **toiletries lying on top of it** in a single line running near to far

along that wall. They are of mixed shapes and sizes. Each one is closed, is dry on the outside, has a **straight middle section a gripper can close across**, and lies still on the pouch without rolling. They lie clear of one another, so each comes off on its own and the pouch is uncovered once they are all gone. Nothing is packed on top of them, none rests on another, and the line is short enough that every one of them fits on the caddy floor.

7. The front center of the case holds the **tote**, sitting upright on the case floor, mouth up and open. It keeps its shape when set down, has no fastening and nothing across its mouth, and its rim is firm enough for a gripper to close on. **Both shoes are inside the tote**, sole down, toes toward the front edge, side by side and not touching, neither on top of the other, and both sitting below the rim so the whole rim is clear.
8. The three garment spots, the toiletry caddy, the pouch spot, and the shoe mat are all empty.

Workspace layout

- **Case:** lying flat at the back of the tabletop, centered between the two arms, base down and squared to the front edge. Nothing holds it in place. A gripper anchors it for the zip, nothing holds it while the lid is opened, every item is lifted straight up so nothing drags it, and it stays lying flat where it was set for the whole episode.
- **Lid rest:** a low block behind the case. The open lid falls back onto it and stays clear of the case mouth for the rest of the episode.
- **Garment spots:** three clear areas of bare tabletop in a row down the right side, within reach of the right arm. They are not marked. The **t-shirt spot** is nearest the case, the **pants spot** sits beyond it, and the **shorts spot** is nearest the front edge, far enough apart that no garment touches another.
- **Toiletry caddy:** on the left, within reach of the left arm. One open caddy with a flat floor and low walls, no slots and no dividers, big enough to hold every toiletry in the case at once, side by side on its floor. Nothing lies across its top, and any handle stands clear of the opening so a gripper lowers straight in. It sits where it is set for the whole episode and is never lifted, carried, or taken by its handle.
- **Pouch spot:** a marked square on the left between the caddy and the front edge, within reach of the left arm. The pouch waits here while the shoes come out of the tote.
- **Shoe mat:** at the front center, a flat mat marked into a left half and a right half, set so the middle line of the mat sits on the middle line of the cell. The left arm works the left half and the right arm works the right half.

The contents are packed on the side they leave toward: garments in the right half of the case for the right arm, the toiletries and the pouch in the left half for the left arm. No carried item passes over another item or over a stack still standing in the case. The tote at the front center and the shoe mat at the front center are the only places both arms work, and each arm keeps to its own side of them.

Vocabulary

- **Arm assignment:** the right arm works the case's right half, the lid, the garment spots, and the right half of the mat. The left arm works the case's left half, the toiletry caddy, the pouch spot, and the left half of the mat. Neither arm takes the other's zones. Two things are shared. The zip is split by side: the right gripper draws the right slider, the left gripper draws the left slider, and whichever one is not on a tab anchors the case. The tote's mouth is taken in turn: whichever gripper is not lifting a shoe or the pouch holds the mouth.
- **Stack:** two or more items resting on top of one another in the case. A stack is worked from the top down, one item at a time, and the only item ever taken is the one with nothing on top of it. Whatever lies under it is left where it is until its turn.
- **Track:** the line of teeth the sliders run along. It has a **middle park** at its center, where both sliders sit together while the case is closed, a **left end stop**, and a **right end stop**. Opening the case means driving the two sliders apart, each out to the end stop on its own side.
- **Loop tab:** the fabric loop on a slider. Each slider has one, and it is the only part of a zip a gripper ever touches.
- **Lid corner:** the front right corner of the lid. It is the only part of the lid a gripper ever lifts by, and only the right gripper ever lifts it.
- **Anchor:** a closed gripper pressed down on the case on its own side, holding the case down and square on the tabletop while the other gripper draws a slider. It goes either on the **side panel** on its own side, level with the middle of the case, or on the **flat top face of the closed lid** on its own side, whichever the arm reaches cleanly. The left gripper anchors while the right slider is drawn and the right gripper anchors while the left slider is drawn. An anchor never travels and never sits on the front face where the sliders run. The zip draws are the only time the case is anchored. Apart from those two anchors, no gripper ever grips, steadies, pushes, or lifts the case.
- **Stand off:** the left gripper holding still where it is, clear of the case and clear of the lid's swing, while the right gripper opens the lid. It does not go home, does not touch the case or the lid, and does not steady anything.
- **Draw:** moving one slider from the middle park out to the end stop on its own side, along the line of the track. **A draw is not one continuous pull.** The arm cannot travel the whole track in one motion, so a draw is worked in as many pulls as it takes: pinch the loop tab, pull along the teeth as far as the arm goes, release, come back down on the same tab where it now sits, and pull again. Every pull runs onward toward the end stop and none runs back toward the middle. There are two draws in an episode, one per slider, and neither gripper ever crosses the middle of the track.
- **Lift:** taking an item straight down onto its grasp, closing, and raising it straight up until there is daylight under it, before anything moves sideways.
- **Load:** carrying an item level to its place, lowering it straight down until it is resting, and releasing.

- **Mouth hold:** a gripper closed across the **middle of the tote's mouth rim**, holding the tote steady and its mouth open while the other gripper lifts a shoe out or lowers the pouch in. A mouth hold is held for the whole of the lift it covers, it does not lift or drag the tote, and it comes off only once the item it covered is clear of the rim or resting on the tote floor.
- **Nest:** the pouch resting on the floor of the tote with no part of it standing at the mouth.
- **Toiletries:** the items packed in the case's left half, whatever they happen to be and however many of them there are. They are worked and named by their place in the line — the **nearest**, then the next one along it, and so on to the **far** one — never by what each item is. The line is worked near to far and is finished when the pouch under it is uncovered. The same items need not be used from one episode to the next.
- **Garment types:** the garments are named **t-shirt**, **pants**, and **shorts**, never by color or pattern. There is one garment of each type, and each type has one spot and only that spot.
- **Adjust:** straightening a garment on its own spot after it has been released, either by nudging or sliding it flat and square with the gripper closed, or by lifting it clear and lowering it again. Garments are the only thing ever adjusted, only on their own spot, only by the right gripper, and only before the next garment is lifted.
- **Settled:** the item stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, or falling over.
- **Release point:** over the garment spot, the pouch spot, the caddy, the mat half, or the tote the item is going into, once the item is already resting. A gripper opens nowhere else.

Steps

Steps 1 to 7 are one suitcase. Every item is lifted straight up out of the case or the tote and carried in the orientation it came out in. Where items are stacked, the top one is taken first and everything under it is left untouched until its own turn comes.

Step 1: Draw the zip open

Goal: both sliders at their end stops and the track open along its whole length, with the case still lying flat and square where it started.

Both sliders start at the **middle park**. The **right gripper draws the right slider out to the right end stop first**, then the **left gripper draws the left slider out to the left end stop**. Each gripper works only the slider on its own side, and the other gripper is holding the case down the whole time it does. Neither gripper ever crosses the middle of the track.

A draw is worked in as many pulls as the arm's travel allows. Pulling a tab, releasing it, coming back down on it and pulling it onward again is how the slider is expected to reach its end stop, and it is not a fault. What is a fault is a slider that travels back toward the middle, comes off the line of the track, or is left short of its end stop.

1.1 Right gripper draws the right slider out

- With the **left gripper** closed, press down on the case on the left side, either on the **left side panel** level with the middle of the case or on the **top face of the closed lid** on that side, and hold it there for the whole draw.
- With the **right gripper**, come down onto the **right slider's loop tab** from above the track, where the two sliders are parked at the middle, and pinch it. Take the right tab only, without touching the left slider or its tab.
- Pull that slider along the track, middle to right, as far as the arm travels. If it is not at the end stop, release the tab, come back down on the same tab where the slider now sits, and pull it onward again. Repeat until the slider reaches the right end stop and will not travel further, then release the tab and lift clear.
- Every pull runs toward the right end stop. Never pull the slider back toward the middle to take a fresh hold.
- The **left gripper** lifts clear only once the right gripper is off the track.

1.2 Left gripper draws the left slider out

- With the **right gripper** closed, press down on the case on the right side, either on the **right side panel** level with the middle of the case or on the **top face of the closed lid** on that side, and hold it there for the whole draw.
- With the **left gripper**, come down onto the **left slider's loop tab**, still parked at the middle, and pinch it.
- Pull that slider along the track, middle to left, taking the tab again and pulling onward as many times as it takes, until the slider reaches the left end stop and will not travel further. Release the tab and lift clear.
- The **right gripper** lifts clear only once the left gripper is off the track.

Each draw pulls the case toward the side it runs to and nothing else holds it, so the anchor is the only thing keeping it square on the tabletop. The anchor sits on the opposite side of the case from the slider being drawn, so the two grippers never meet on the front of the case.

Check: one slider sits at the right end stop, the other at the left end stop, the track is open along its whole length, and the case has not slid or lifted off the tabletop. If a slider stopped short, take its loop tab again with the same gripper and pull it the rest of the way.

Step 2: Open the lid

Goal: the lid lying back on the lid rest, with the whole case interior open to view.

The **right gripper opens the lid on its own**. The left gripper takes no part in this step.

- The **left gripper** stands off: having lifted clear of the track at the end of Step 1, it holds still there, off the case and clear of the path the lid swings through, until the step is over. It does not return home, and it does not touch the case, the lid, or anything in the case.

- The **right gripper** comes down onto the **front right corner of the lid** from above and closes across it.
- Lift that corner up and back over the hinge, following the arc the hinge swings through, until the lid has passed upright and is falling back toward the lid rest on its own.
- **Release it there, in the air.** The arm does not reach all the way to the lid rest, so once the lid is past upright and falling backward it is let go and drops the rest of the way onto the rest. Releasing before it is past upright is the fault, not releasing in the air.
- Lift clear once the lid is down.

The lid is taken by one corner and not by the middle, so the right arm never has to reach across the open mouth.

Check: the lid rests on the lid rest, the case mouth is clear from front to back, and the case has not slid or lifted off the tabletop. If the lid has fallen back toward the case or landed beside the rest, take the same corner again and carry it back past upright once more.

Step 3: Lay the garments out

Goal: the case's right half empty and each garment lying flat on the spot for its type.

Work by type in this order: the t-shirt, then the pants, then the shorts. That is the order they are stacked in, so the garment being taken is always the one lying on top of the stack.

- The **right gripper** closes across the **left side of the garment** on top of the stack, on the folded edge that runs down that side, taking the full thickness of that garment's fold and no part of the one under it, and lifts it straight up off the stack until there is daylight under it.
- Carry it level to its spot: the t-shirt to the **t-shirt spot** nearest the case, the pants to the **pants spot** beyond it, the shorts to the **shorts spot** nearest the front edge.
- Lower it flat onto its spot, squared up the way it came out, and release once it is resting.
- **Adjust it if it needs it.** A garment that lands crooked, rucked, or off its spot may be straightened where it lies: nudge or slide it flat and square with the gripper closed, or lift it clear and lower it again. Adjust only the garment just placed and only on its own spot, and stop once it lies flat.
- Move to the next type only once the current garment is down on its spot and any adjusting is finished. Once the next garment is lifted, the one before it is not touched again.

Take each garment straight up off the top of the stack. Never slide one out from under the garment above it, never lift two at once, never shake a garment out to check what it is, and never refold one once it is down.

Check: the case's right half is empty, the three garments lie flat on their spots in the order t-shirt, then pants, then shorts running from the case to the front edge, and no garment touches another or overlaps one. If a garment lands on the wrong spot or on top of another, lift it clear and lower it into place again. Whatever is still in the case is left exactly as it lies.

Step 4: Move the toiletries into the caddy

Goal: every toiletry resting in the toiletry caddy, the caddy holding nothing else, and the pouch uncovered in the case.

The toiletries are packed on top of the pouch and come out before it. Each one is taken off the pouch on its own and the pouch is left where it lies.

- The **left gripper** closes on the toiletry nearest the front edge **across the middle of its body**, not on a cap, a lid, or either end, and lifts it straight up off the pouch until there is daylight under it, taking that toiletry alone and no part of the pouch under it.
- Carry it level to the **toiletry caddy**, hold it inside the caddy walls, lower it straight down until it rests on the caddy floor, and release.
- Take the next toiletry in the line the same way, and go on down the line, each one into a clear part of the caddy floor beside the ones already there, until nothing is left on the pouch.
- Leave the pouch exactly as it lies. If it has rucked or shifted when a toiletry lifted, it is taken as it lies in Step 5, not straightened now.

Work the toiletries near to far, the order they are packed in. Do not sort them by size and do not go back to a toiletry once you have moved on.

Check: every toiletry rests in the caddy, none is on a caddy wall, none hangs over the rim, none is stacked on another, all are settled, and the pouch lies uncovered on the floor of the case's left half and has not been lifted or carried up with a toiletry. If a toiletry sits on the rim or on top of another, lift it clear and lower it into a free part of the floor.

Step 5: Take the pouch out onto the pouch spot

Goal: the pouch lying flat on the pouch spot and the case's left half empty.

The pouch is uncovered once every toiletry is in the caddy. It goes to the pouch spot first and only reaches the tote at the end of the episode, once the shoes are out of it.

- The **left gripper** closes on the pouch across its **middle** and lifts it straight up off the case floor until there is daylight under it.
- Carry it level to the **pouch spot**, lower it flat onto the mark, and release once it is resting.

Check: the pouch lies flat on the pouch spot and the case's left half is empty. If the pouch lands off its spot, lift it clear and lower it onto the mark again.

Step 6: Take the shoes out of the tote and onto the mat

Goal: both shoes sitting sole down on the mat, toes toward the front edge, side by side and not touching, and the tote standing empty and upright where it was packed.

The tote stays in the case for the whole step. It is never lifted, never dragged, and never tipped to get a shoe out. One gripper holds its mouth while the other lifts, and **the two arms switch**

after the left shoe: the left shoe is lifted by the left gripper, the right shoe by the right gripper, so neither arm reaches across the tote.

6.1 Left shoe

- The **right gripper** closes across the **middle of the tote's mouth rim** and holds it there, steady, for the whole of this lift. It holds the tote down and its mouth open; it does not lift it, pull it forward, or move it on the case floor.
- The **left gripper** closes on the **left shoe** across its heel, taking both sides of the heel counter, and lifts it straight up out of the tote, level and sole down, until the whole shoe is clear of the rim.
- Carry it to the **left half of the mat**, lower it until the whole sole is resting flat, and release.
- The **right gripper** comes off the rim only once the shoe is clear of the tote.

6.2 Switch, then the right shoe

- The arms swap roles. The **left gripper** now closes across the **middle of the tote's mouth rim** and holds it steady.
- The **right gripper** closes on the **right shoe** across its heel and lifts it straight up out of the tote, level and sole down, until the whole shoe is clear of the rim.
- Carry it to the **right half of the mat**, lower it until the whole sole is resting flat, and release.
- The **left gripper** comes off the rim only once the shoe is clear of the tote.

Both shoes are packed sole down and facing the front edge, so neither one is turned at any point. Carry each one the way it came out.

Check: both shoes sit flat on the mat, soles down, toes toward the front edge, one on each half, side by side and not touching; the tote stands upright and empty in the front center of the case, where it was packed. If a shoe lands on its side or overhangs the mat, lift it clear and set it down again.

Step 7: Nest the pouch in the tote

Goal: the pouch resting inside the tote, with the tote still standing in the case.

- The **right gripper** takes the **mouth hold** again at the middle of the tote's mouth rim and holds it there for the whole step.
- The **left gripper** closes on the pouch on the **pouch spot** across its middle, lifts it straight up until there is daylight under it, carries it level over the open mouth, lowers it straight down until it rests on the tote floor, and releases.
- Both grippers lift clear.

The tote is emptied of the shoes before the pouch goes in, so nothing is ever pushed down onto something already inside it.

Check: the pouch is fully inside the tote with no part of it standing at the mouth, the tote is still upright in the front center of the case and has not moved, the pouch spot is empty, and both halves of the case are empty. If the pouch stands proud of the mouth, take the mouth hold again, lift the pouch out, and lower it back in.

Step 8: End the episode

- Confirm the case holds nothing but the tote with the pouch inside it, the lid is back on the lid rest, the three garments lie flat on their spots, the toiletries rest in the caddy, both shoes sit on the mat, and the pouch spot is empty.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the pouch out of the tote and set it aside.
2. Put both shoes back inside the tote, sole down, toes toward the front edge, side by side and not touching, both below the rim. Leave the tote standing upright in the front center of the case, mouth up and open, and the shoe mat empty.
3. Lay the pouch flat on the floor of the case's left half, square to the left wall. The pouch spot is left empty.
4. Take the toiletries out of the caddy and lay them back on top of the pouch in a single line running near to far along the case's left wall, each one closed, clear of one another and with nothing on top of them. Leave the caddy empty.
5. Rebuild the garment stack in the case's right half: the shorts on the case floor, the pants on it, the t-shirt on top, each squared up face up so none overhangs the one below. Refold any garment that has come loose so it is a flat rectangle with a clean folded edge down its left side.
6. The stack order never varies, because it is the order the garments are worked in. Vary where the garment stack, the toiletries, and the tote sit within their parts of the case between episodes rather than rebuilding the same layout every time.
7. Lift the lid off the rest, lay it down onto the case, and draw both sliders back along the track until they meet at the middle park so the case is closed.
8. Square the case at the back of the tabletop: base flat, front edge parallel to the front edge of the tabletop, lid facing up.
9. Check both loop tabs are attached, the lid's front right corner is sound, the lid's top face is firm where the anchor presses, and both sliders run freely between the middle park and their end stops. A zip that is starting to catch is replaced before the next episode.
10. Replace any item that is damaged: a case that will not sit square, a garment that will not hold a fold, a toiletry that leaks or has lost its cap, a shoe that will not sit flat, or a tote that

no longer holds its shape or whose rim will not take a hold.

11. Wipe the tabletop clear of the three garment spots and confirm the lid rest, the caddy, the pouch spot, and the mat sit firm and have not shifted.
12. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** the left slider is drawn before the right one is at the right end stop, the lid is lifted before both sliders are at their end stops, an item is taken out before the lid is on its rest, the pouch is lifted before every toiletry is in the caddy, a shoe is lifted before the pouch is on the pouch spot, or the pouch goes into the tote before both shoes are on the mat.
- **SOP rule broken:** Steps 1 to 7, the order never changes and a zone is finished before the next one starts.
- **Coaching note:** finish the zone you are in before you start another.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes a garment or the lid; the right gripper reaches the toiletry caddy, the pouch spot, or the case's left half; the left gripper lifts the right shoe or the right gripper lifts the left shoe; or a shoe is set on the wrong arm's half of the mat by the arm that lifted it. Each gripper taking the slider on its own side in Step 1, and each gripper taking the tote's mouth in its turn, are not this violation.
- **SOP rule broken:** Steps 1 to 7, the right arm works the lid, the case's right half, the garment spots, and the right shoe; the left arm works the case's left half, the caddy, the pouch spot, and the left shoe; each arm works only the slider on its own side; and the arms switch on the tote's mouth rather than reaching across it.
- **Coaching note:** each arm stays on its own zones. Left shoe left arm, right shoe right arm, and the other gripper takes the mouth.

Violation: Zip drawn without an anchor

- **Visible cue:** no gripper is on the case while a slider travels and the case twists, shifts, or is dragged across the tabletop; the anchor comes off the case between pulls or before the drawing gripper is clear of the track; an anchor sits on the front face on or beside the track instead of on the side panel or the lid's top face; or an anchor travels along with the tab.
- **SOP rule broken:** Step 1, nothing but the anchor holds the case, so the gripper that is not on a tab holds the case down on its own side — on the side panel or on the top face of the closed lid — for the whole of the draw, including between pulls.
- **Coaching note:** anchor first, then pull, and keep the anchor down while you take the tab again. The anchor belongs on the side panel or on the lid, never on the zip line.

Violation: Zip drawn wrong or left short

- **Visible cue:** a slider is pulled forward off the case, lifted off the fabric, drawn in an arc rather than along the teeth, or driven back toward the middle to set up another pull; or a draw finishes short of its end stop with the case still fastened along part of the track and the episode moves on. Releasing the tab and taking it again to pull onward is not this violation.
- **SOP rule broken:** Step 1, a draw runs along the line of the track in as many pulls as it takes, every pull onward toward the end stop, from the middle park out to the end stop on its own side.
- **Coaching note:** as many pulls as you need, all of them forward. Look at the end stop before you move on.

Violation: Wrong slider, wrong gripper, or both at once

- **Visible cue:** a gripper takes the slider on the other side of the middle park; a gripper pinches both tabs at once or catches the second tab while drawing the first; one gripper draws both sliders; or both sliders travel at the same time.
- **SOP rule broken:** Step 1, the right gripper draws the right slider and the left gripper draws the left slider, one at a time, and neither gripper crosses the middle of the track.
- **Coaching note:** one tab, one gripper, one draw. Take the tab on your own side and leave the other parked until its turn.

Violation: Only one slider drawn

- **Visible cue:** the episode moves on to the lid with one slider still at the middle park, or the second draw is skipped entirely and the case is opened against a track that is still fastened on one side.
- **SOP rule broken:** Step 1, both sliders are drawn out to their end stops before the lid is touched.
- **Coaching note:** two sliders, two draws. Check both end stops before you go near the lid.

Violation: Lid forced, released early, or left off the rest

- **Visible cue:** the lid is pushed, shoved, or levered open by the shell, or taken by the middle of its front edge or by any corner other than the front right one; the right gripper releases it before it has passed upright and it falls back onto the case; or it lands beside the lid rest, on the tabletop, or back on the case and is left there. Releasing the lid in the air once it is past upright and falling toward the rest is not this violation.
- **SOP rule broken:** Step 2, the right gripper closes on the lid's front right corner, carries it up and back along the hinge's arc until it is past upright and falling toward the rest, and releases it there so it drops onto the lid rest.
- **Coaching note:** take the right corner, carry it past upright, then let it go. Past upright before you release, and on the rest before you move on.

Violation: Left gripper worked during the lid

- **Visible cue:** the left gripper anchors, steadies, pushes, or pins the case while the lid is opened; takes the lid or helps carry it; returns home mid-step; or drifts into the path the lid swings through.
- **SOP rule broken:** Step 2, the right gripper opens the lid on its own and the left gripper stands off, holding still and clear until the step is over.
- **Coaching note:** the lid is a one-arm job. The left gripper waits where it is and does nothing.

Violation: Case lifted or moved

- **Visible cue:** the case slides, rotates, or tips away from where it was set, is lifted off the tabletop, or is gripped, steadied, pushed, or pinned by either gripper anywhere other than that gripper's anchor on its own side during a zip draw.
- **SOP rule broken:** Steps 1 to 7, the case stays lying flat where it was set, and the two zip anchors are the only holds ever taken on the case.
- **Coaching note:** anchor for the zip draws only, and lift straight up so nothing drags the case. If the case has travelled, stop and set it flat and square again.

Violation: Tote lifted, dragged, or tipped

- **Visible cue:** either gripper lifts the tote off the case floor, pulls it forward or sideways across the floor, tips or tilts it to shake a shoe loose, takes it by its body or its base instead of the middle of its mouth rim, or carries it out of the case to any zone.
- **SOP rule broken:** Steps 6 and 7, the tote is worked where it stands in the case, held only by the middle of its mouth rim, and it is still upright in the front center when the episode ends.
- **Coaching note:** the tote does not travel. Hold the rim, lift the shoe out of it, and leave the tote where it sits.

Violation: Item taken from under another

- **Visible cue:** a gripper closes on an item that still has another item resting on it, slides one out from beneath the item above it, lifts two items at once, or moves an item aside to reach another; or the items left under the lift are dragged, tipped, or carried up with it — the pouch lifting with a toiletry, or a garment lifting with the one above it.
- **SOP rule broken:** Steps 3 and 4, every stack is worked from the top down, one item at a time, and nothing under the top item is touched.
- **Coaching note:** the next item is always the one on top. Come down on it, lift it straight off, and leave the rest of the stack where it is.

Violation: Item taken by the wrong part

- **Visible cue:** the right gripper takes a garment by a loose corner, a sleeve or a leg, or a single leaf of the fold rather than across its left side, or takes the lid by anything but its front right corner; either gripper takes a slider by its body, the teeth, the zip tape, or the fabric beside the track rather than by its loop tab, or takes the tote by anything but the middle of its mouth rim; the left gripper takes a toiletry by its cap, its lid, or either end, or the pouch by a corner, rather than across its middle; or a shoe is taken by its laces, tongue, or sole rather than across the heel.
- **SOP rule broken:** Steps 1 to 7, each item has one grasp.
- **Coaching note:** land on the grasp point or reset and take it again.

Violation: Item dragged instead of lifted

- **Visible cue:** an item slides along the case floor, the tote floor, or the tabletop still touching it, with no daylight under it in transit; or a shoe is dragged up the inside wall of the tote instead of being lifted clear of the rim.
- **SOP rule broken:** Steps 1 to 7, the gripper lifts straight up until there is daylight before moving sideways.
- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Item turned or regripped in the air

- **Visible cue:** the right gripper flips, swings, or shakes out a garment during the carry; either gripper turns the pouch over; a shoe is rolled off its sole; or a gripper shifts, rolls, or seats its hold again on an item in the air without setting it down first. Taking the loop tab again between pulls in Step 1, and lifting a placed garment clear to lay it again in Step 3, are not this violation.
- **SOP rule broken:** Steps 2 to 7, nothing is turned in the air, each item is carried in the orientation it came out in, and it is set down before a new hold is taken.
- **Coaching note:** orientation is set at the pick, never in the air. Put it down before you take a new hold.

Violation: Garment on the wrong spot or not flat

- **Visible cue:** the right gripper lays the t-shirt, the pants, or the shorts on another type's spot, lays two garments on one spot or touching, works the types in any order other than t-shirt, then pants, then shorts, or leaves a garment folded over on itself, rucked, or overlapping another when the next garment is lifted.
- **SOP rule broken:** Step 3, each garment type has one spot, the types are worked in a fixed order, and each garment is left lying flat and square on its own spot before the next one is lifted.
- **Coaching note:** name the type before you lift it, lay it flat on its own spot, and straighten it before you move on.

Violation: Garment touched after its turn, or a placement adjusted that is not a garment

- **Visible cue:** the right gripper goes back to a garment already finished, once the next garment has been lifted; a garment is refolded, patted, or squashed rather than nudged flat; or a gripper nudges, slides, or straightens anything that is not a garment on its spot — a toiletry in the caddy, a shoe on the mat, the pouch, or the tote.
- **SOP rule broken:** Step 3, garments are the only thing adjusted, only on their own spot, and only before the next garment is lifted; everywhere else a bad placement is corrected by lifting the item clear and lowering it again.
- **Coaching note:** straighten the garment you just laid, then leave it. Nothing else gets nudged.

Violation: Pouch not staged on the pouch spot

- **Visible cue:** the pouch goes straight from the case into the tote instead of onto the pouch spot; it is left on the tabletop, on the pouch spot, or anywhere but inside the tote at the end of the episode; or it is moved to the tote before both shoes are on the mat.
- **SOP rule broken:** Steps 5 and 7, the pouch comes out onto the pouch spot once every toiletry is in the caddy and goes into the tote only after both shoes are on the mat.
- **Coaching note:** toiletries first, then the pouch out to its spot, shoes out of the tote, pouch in last.

Violation: Toiletry not in the caddy

- **Visible cue:** the left gripper leaves a toiletry on a caddy wall, hanging over the rim, on top of another toiletry, or on the tabletop beside the caddy; it works the toiletries in any order other than near to far; or either gripper lifts, drags, or takes the caddy by its handle instead of leaving it where it stands.
- **SOP rule broken:** Step 4, every toiletry rests on the caddy floor, the toiletries are worked near to far, and the caddy itself stays where it is set.
- **Coaching note:** inside the walls, down on the floor, next to the ones already there. The caddy does not move.

Violation: Shoes wrong side or not flat

- **Visible cue:** the left shoe is set on the right half of the mat or the reverse, a shoe is set down on its side or on its toe, the toes point away from the front edge, the two shoes are left touching, or a shoe overhangs the mat.
- **SOP rule broken:** Step 6, both shoes sole down, toes toward the front edge, one on each half of the mat, side by side and not touching.
- **Coaching note:** left shoe left half, right shoe right half, both flat.

Violation: Tote worked without a mouth hold

- **Visible cue:** a shoe is lifted out of the tote, or the pouch is lowered into it, with no gripper on the mouth rim; the mouth hold is taken only after the shoe or the pouch is already in the air; the holding gripper comes off the rim before the shoe is clear of it or before the pouch is resting on the tote floor; or the holding gripper pulls the tote about instead of holding it steady.
- **SOP rule broken:** Steps 6 and 7, one gripper holds the middle of the mouth rim before the other one takes hold of the shoe or the pouch, and keeps holding until that item is clear of the rim or resting inside.
- **Coaching note:** hold the rim first, let go last.

Violation: Item not lowered onto its place

- **Visible cue:** a gripper opens above a garment spot, the pouch spot, the caddy, the mat, or the tote mouth and the item falls the rest of the way; a gripper pushes an item in sideways instead of lowering it; a gripper presses, pats, or squashes the contents of the tote or the items still in the case; or a gripper opens over the case, over another zone, or over open space. The lid released in the air in Step 2 is not this violation.
- **SOP rule broken:** Steps 3 to 7, a gripper opens over the place the item is going into, only once the item is resting, and a bad placement is corrected by lifting the item clear and lowering it again.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Item dropped or something knocked over

- **Visible cue:** an item falls onto the tabletop or the floor, either gripper strikes the case and it slides out of place, a stack still in the case is knocked over or spread across the case floor, the caddy, the pouch spot marker, or the mat is pushed out of its spot, a toiletry is knocked over, the tote is knocked over inside the case, the lid is knocked off its rest, a garment already down is dragged out of place, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 7, each item stays on a clear travel path, nothing already finished is touched again, and a gripper releases only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a garment, a toiletry, or a shoe still in the case or the tote, a garment spot empty, a toiletry outside the caddy, a shoe off the mat, the pouch still on the pouch spot or anywhere but inside the tote, the tote out of the case, or either arm away from home.
- **SOP rule broken:** Step 8, confirm the case holding nothing but the tote with the pouch inside it and all four loaded zones, return both arms home, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Zip jammed or a slider derailed** under pulls along the line of the track, or the two sliders will not separate at the middle park. Replace the case before the next episode.
- **A loop tab missing or torn off**, or the lid's front right corner split or crushed so a gripper cannot close on it, when the step begins.
- **Lid will not stay on the lid rest:** it bounces off the rest when released past upright, or falls back onto the case under its own weight. Reseat or replace the rest before the next episode.
- **Defective item:** a garment that comes apart in the gripper and will not hold a fold, a leaking toiletry or one that has lost its cap, a shoe that will not sit flat, or a tote that collapses, folds shut, or whose rim will not hold its shape under a mouth hold.
- **Lid rest, caddy, pouch spot, or mat shifts out of its spot** during the episode.
- **The case will not stay put:** it slides across the tabletop while an anchor is held on the side panel or the lid, or its base is warped and it will not sit flat. Replace the case before the next episode.
- **The case will not hold its contents at reset:** a stack will not stand square inside it, the tote will not stand upright in the front center, or the lid will not close over the packed contents. Replace the case before the next episode.
- **The toiletries will not all sit side by side** on the floor of the caddy, or **both shoes will not sit side by side inside the tote** below its rim.

Annotation subtasks (from SOP)

1. Anchor the case for a zip draw

2. Draw one slider from the middle park out to the end stop on its own side
3. Open the case lid onto the lid rest
4. Move one garment from the top of the stack onto its spot
5. Straighten a garment on its spot
6. Move one toiletry into the caddy
7. Move the pouch from the case onto the pouch spot
8. Hold the tote steady by the middle of its mouth rim
9. Take one shoe out of the tote and set it on the mat
10. Nest the pouch inside the tote
11. Return both arms home and end the episode